

# Quantum Molecular Encoding Captures Stereochemical Complexity of African Natural Products for Antimalarial Drug Discovery

Myke Vital Sao Temgoua <sup>11</sup>, Jean-Pierre Tchapel Njafa <sup>1</sup>, Penabei Samafou <sup>2</sup>, Fon Wilfred Mbacham <sup>3</sup>, Serge Guy Nana Engo <sup>1</sup>

<sup>1</sup>Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

<sup>2</sup>Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke, Sherbrooke, QC, Canada

<sup>3</sup>Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812, Yaoundé, Cameroon

## Abstract

[PENDING: Abstract: Write last after Results are complete. Target: 200 words. Structure: (1) Challenge: ANP stereochemical complexity + classical fingerprint limitations; (2) Approach: BondOrderMatrix quantum encoding + quantum kernel SVM on 17 ANP-derived candidates; (3) Results: quantum vs. ECFP4 comparison with statistical test; (4) Interpretation: advantage/equivalence/underperformance + mechanistic analysis; (5) Scope: Phase 1 proof-of-concept, honest-negative framework.]

**Keywords:** quantum machine learning, African natural products, antimalarial drug discovery, drug resistance, molecular encoding, quantum kernels, stereochemistry, BondOrderMatrix

## 1 Introduction

Antimalarial drug discovery faces a dual challenge: the *Plasmodium falciparum* parasite’s capacity for rapid resistance evolution against single-target therapeutics, and the structural complexity of African natural products (ANPs) that have historically yielded effective but under-exploited antimalarials. Artemisinin from *Artemisia annua* and quinine from *Cinchona* bark exemplify this paradox—ANP-derived drugs provide clinical benefit yet their polycyclic, stereochemically rich frameworks resist translation into scalable synthetic libraries[1]. African ethnobotanical sources including *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia* contain structurally diverse alkaloids, prenylated flavonoids, and quassinoids with multitarget activity profiles[2], yet computational screening pipelines default to two-dimensional molecular fingerprints that discard the three-dimensional stereochemical information critical to ANP bioactivity.

Classical cheminformatics encodes molecules as fixed-length bit vectors (e.g., ECFP4[3]) or graph message-passing representations (graph neural networks, GNNs[4]). While effective for flat, aromatic drug-like molecules, these methods exhibit systematic blind spots for ANP structural motifs: ECFP4 radius-based hashing collapses stereoisomers into identical bit patterns, and k-hop message passing in GNNs cannot distinguish cis/trans double bonds, fused ring geometries, or axial/equatorial substituent configurations[5]. For molecules with high fractions of sp<sup>3</sup> carbons ( $F_{sp^3} > 0.5$ ), multiple stereocenters, and flexible polycyclic frameworks—hallmarks of ANP chemistry—this loss of three-dimensional information directly degrades predictive accuracy. Prior work in this project series demonstrated 87.5% estimand divergence between static docking and molecular dynamics simulations for ANP-like candidates[6], attributed to conformational flexibility and solvation effects unaccounted for by rigid 2D fingerprints. Separately, scaffold-split benchmarks showed GNN underperformance (AUC = 0.649) relative to ECFP4-Random Forest

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<sup>1</sup>Corresponding author: myke-vital.sao@facsciences-uy1.cm

(0.689) on the same ANP-enriched dataset[7], consistent with the 1-Weisfeiler-Lehman (1-WL) expressivity bottleneck[5] failing to generalize across complex polycyclic scaffolds.

Quantum machine learning (QML) offers a fundamentally different encoding strategy: molecules are represented as quantum states in a  $2^n$ -dimensional Hilbert space, where entanglement can capture correlations between distant atoms—critical for long-range electrostatics, ring strain, and stereoelectronic effects in ANP structures. Quantum kernel methods[8] compute similarity between quantum states via inner products, yielding kernel matrices that preserve molecular geometry without dimensionality reduction. Recent demonstrations of quantum advantage in chemical property prediction have emerged in specialized contexts[9, 10], yet systematic benchmarks on stereochemically complex natural products remain absent from the literature. Crucially, prior quantum approaches that first reduce molecules to classical fingerprints before quantum encoding[11] necessarily inherit the same stereochemical blind spots, limiting quantum enhancement to kernel structure rather than molecular representation.

Here we investigate whether quantum molecular encoding preserves African natural product structural complexity sufficiently to improve antimalarial activity prediction relative to classical baselines. We encode 17 candidate molecules from a curated ANP-derived panel (sourced from the upstream P1 library of 65 856 computationally expanded structures[12]) using BondOrderMatrix representations[9], where bond orders, Z/E stereochemistry, and R/S chirality are embedded directly into parameterized quantum circuits. Classification via quantum kernel support vector machines is compared against ECFP4-SVM under leave-one-out cross-validation on an intentionally small, class-imbalanced cohort designed to stress-test generalization to unseen scaffolds. This proof-of-concept Phase 1 study addresses three questions: (i) Does quantum encoding capture ANP stereochemistry that ECFP4 discards? (ii) Does this encoding translate to measurable classification advantage, or do quantum and classical kernels yield equivalent performance on real-world antimalarial screening data? (iii) For which ANP structural classes—indole alkaloids, prenylated flavonoids, polycyclic quassinoids, or simple phenolics—does quantum enhancement provide advantage versus equivalence?

Following the transparent reporting precedent established in prior quantum-inspired benchmarks[11], we adopt an honest-negative framework: statistical equivalence or quantum underperformance are reported with identical rigor to quantum advantage, accompanied by mechanistic analysis of when and why quantum encoding succeeds or fails. This approach prioritizes falsifiable claims over promotional framing, aligning with recent calls for reproducible quantum machine learning benchmarks in drug discovery[13].

## 2 Methods

### 2.1 Dataset: African Natural Product-Derived Antimalarials

We extracted 17 antimalarial candidate molecules from Project 1 Set A[12], a curated panel enriched for African natural product (ANP) structural motifs prioritized by multi-parameter optimization ( $MPO \geq 0.70$ ) and Resistance-Resilience Score (RRS) across four malaria drug targets (PfDHFR, PfCRT, PfATP4, PfClpP). The seed library was derived from 396 ANP structures sourced from ANPDB[2] and AfroDb[? ], encompassing traditional antimalarial botanical species including *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia*. All molecules were represented as canonical SMILES strings and validated using RDKit[14].

#### 2.1.1 African-Chemotype Characterization

To quantify ANP structural identity, we computed three descriptors for each molecule:

(i) **Fraction  $sp^3$  carbons ( $F_{sp^3}$ )**: Calculated as the ratio of  $sp^3$ -hybridized carbons to total carbons. ANPs typically exhibit  $F_{sp^3} > 0.5$  due to saturated ring systems and polycyclic frameworks, contrasting with synthetic drugs ( $F_{sp^3} \sim 0.3\text{--}0.4$ )[15].

(ii) **Stereocenter count:** Enumerated using RDKit’s `FindMolChiralCenters` function after stereochemistry assignment. ANP alkaloids, triterpenoids, and macrolides frequently contain 4–10 stereocenters[? ].

(iii) **African-Chemotype Structural Index (ACSI):** Defined as the mean Tanimoto similarity (Morgan fingerprint, radius=2) to the five nearest neighbors within the 396-member ANP reference set. ACSI ranges from 0 (no ANP similarity) to 1 (perfect match); values  $> 0.5$  indicate strong ANP character.

Activity labels (active/inactive) were derived from docking-score thresholds applied in the upstream P1 multi-target triage workflow (see [LED-001] for label assignment criteria and validation). The final panel comprised 15 inactive and 2 active molecules (class imbalance 15:2), creating a challenging benchmark representative of hit-discovery screening where active hits are rare.

## 2.2 Classical Baseline: ECFP4 Fingerprints

Extended-Connectivity Fingerprints (ECFP4)[3] were generated using RDKit (`GetMorganFingerprintAsBitVec` radius=2, 2048 bits). A support vector machine (SVM) with radial basis function (RBF) kernel was trained using scikit-learn (v1.3.0)[16]. Hyperparameters  $C$  and  $\gamma$  were set to default values (1.0 and `scale`, respectively). Performance was evaluated via leave-one-out cross-validation (LOO-CV), where each of the 17 molecules served as a held-out test instance while training on the remaining 16. Predictions were scored using the SVM decision function, and area under the receiver operating characteristic curve (AUC-ROC) was computed as the primary metric.

This baseline represents the standard molecular fingerprinting approach used in P1 scaffold-recovery validation (69.3% scaffold conservation)[12] and P3 quantum-inspired benchmarks (AUC 0.9475)[11].

## 2.3 Quantum Molecular Encoding

### 2.3.1 BondOrderMatrix Representation

Following Boy et al.[9], molecular structures were encoded as BondOrderMatrix representations preserving stereochemistry and bond topology:

$$\mathbf{M}_{ij} = \begin{cases} 0.5 \times Z_i^{2.4} & i = j \text{ (diagonal)} \\ \frac{Z_i \times Z_j}{b_{ij}} & i \neq j \text{ (off-diagonal)} \end{cases} \quad (1)$$

where  $Z_i$  is the atomic number of atom  $i$ , and  $b_{ij}$  is the bond order between atoms  $i$  and  $j$  (1 for single, 1.5 for aromatic, 2 for double, 3 for triple bonds).

**Stereochemical encoding:**

- **Z/E isomers:** Z-configuration bonds assigned negative bond orders ( $b_{ij} < 0$ ).
- **R/S chirality:** S-configuration stereocenters assigned negative diagonal elements ( $\mathbf{M}_{ii} < 0$ ).

This encoding directly embeds 3D stereochemical information into the quantum state, unlike classical fingerprints which treat stereoisomers as distinct bit patterns without spatial relationships.

### 2.3.2 Quantum Circuit: BondFeatureMap

The BondOrderMatrix was mapped to a parameterized quantum circuit using PennyLane (v0.32)[17]:

(i) **Initial encoding layer:**

$$|\psi_0\rangle = \bigotimes_{i=1}^n R_Y(\theta_i)|0\rangle, \quad \theta_i = \mathbf{M}_{ii} \quad (2)$$

(ii) **Entangling layer:**

$$U_{\text{ent}} = \prod_{\langle i,j \rangle} R_{ZZ}(\phi_{ij}), \quad \phi_{ij} = \mathbf{M}_{ij} \quad (3)$$

where  $R_{ZZ}(\phi) = \exp(-i\phi Z \otimes Z/2)$  creates two-qubit entanglement encoding off-diagonal matrix elements.

The circuit depth was set to 1 layer (encoding only) for the proof-of-concept Phase 1 study. The number of qubits was determined by the number of heavy atoms (C, N, O, S, P) in each molecule, resulting in circuits ranging from 12 to 28 qubits for the 17-molecule panel.

### 2.3.3 Quantum Kernel Computation

The quantum kernel between molecules  $A$  and  $B$  was computed as:

$$K(A, B) = |\langle 0|U_B^\dagger U_A|0\rangle|^2 \quad (4)$$

where  $U_A$  and  $U_B$  are the BondFeatureMap circuits for molecules  $A$  and  $B$ , respectively. The kernel represents the squared overlap of quantum states, measuring structural similarity in Hilbert space.

For the  $n = 17$  molecule dataset, the  $n \times n$  kernel matrix  $\mathbf{K}$  was computed using PennyLane’s statevector simulator (`default.qubit` backend). Exploiting kernel symmetry ( $K(A, B) = K(B, A)$ ), only  $n(n + 1)/2$  unique entries were evaluated. For  $n = 17$  molecules, this required 153 kernel evaluations.

Computation time per kernel entry was logged to assess scalability. At approximately 5 min per kernel entry (28-qubit circuits, CPU single-threaded), total computation time was  $\sim 13$  h. Parallelization across 8 cores reduced wall-clock time to  $\sim 1.6$  h. All quantum simulations were performed on [PENDING: [hardware: CPU model, RAM]].

## 2.4 Classification with Quantum Kernel

The quantum kernel matrix  $\mathbf{K}$  was used as a precomputed kernel in scikit-learn’s SVM classifier:

```
SVC(kernel='precomputed', C=1.0)
```

Classification was performed using LOO-CV, matching the ECFP4 baseline protocol. For each fold, the  $(n - 1) \times (n - 1)$  training kernel and  $(n - 1) \times 1$  test kernel were extracted from the full  $n \times n$  matrix.

## 2.5 Statistical Analysis

### 2.5.1 Performance Metrics

Primary: AUC-ROC (accounts for class imbalance).

Secondary: Accuracy, precision, recall, F1-score, confusion matrix.

For the 15:2 inactive:active imbalance, we report balanced accuracy:

$$\text{Balanced Acc} = \frac{\text{Sensitivity} + \text{Specificity}}{2} \quad (5)$$

### 2.5.2 Quantum vs. Classical Comparison

Performance difference was assessed using a paired  $t$ -test on AUC scores across the 17 LOO folds (one per molecule). Effect size was quantified using Cohen’s  $d$ :

$$d = \frac{\text{AUC}_{\text{quantum}} - \text{AUC}_{\text{ECFP4}}}{\sigma_{\text{pooled}}} \quad (6)$$

Statistical significance was declared at  $\alpha = 0.05$ . Following the honest-negative precedent established in Project 3[11], we classify outcomes as:

- **Advantage:**  $\Delta\text{AUC} \geq 0.01$  and  $p < 0.05$
- **Equivalence:**  $|\Delta\text{AUC}| < 0.01$  or  $p \geq 0.05$
- **Underperformance:**  $\Delta\text{AUC} < -0.01$  and  $p < 0.05$

### 2.5.3 Kernel Target Alignment

To assess whether the quantum kernel captures activity-relevant structure, we computed kernel target alignment (KTA)[18]:

$$\text{KTA} = \frac{\langle \mathbf{K}, \mathbf{y}\mathbf{y}^\top \rangle_F}{\|\mathbf{K}\|_F \|\mathbf{y}\mathbf{y}^\top\|_F} \quad (7)$$

where  $\mathbf{y}$  is the label vector and  $\langle \cdot, \cdot \rangle_F$  is the Frobenius inner product. KTA ranges from  $-1$  (perfect anti-correlation) to  $+1$  (perfect alignment); values near 0 indicate no correlation between kernel similarity and label similarity.

## 2.6 Computational Provenance

All analyses were conducted using Python 3.10 in a reproducible conda environment (`malaria_qml_hybrid`). Key package versions: PennyLane 0.32.0, PyTorch 2.0.1, RDKit 2023.09.1, scikit-learn 1.3.0, NumPy 1.24.3, Pandas 2.0.3. Random seeds were fixed (seed=42) for all train/test splits. Quantum circuit parameters, kernel matrices, and SVM models were serialized and archived alongside results. Detailed provenance metadata, including computation timestamps and hardware specifications, are provided in the Supporting Information.

Code and data are available at [PENDING: [GitHub repository or Zenodo DOI upon acceptance]].

## 3 Results

[PENDING: Results section: Waiting for LED-PENDING-001 (17-molecule quantum kernel + SVM classification + statistical comparison). Cannot draft until full quantum results available.]

## 4 Discussion

[PENDING: Discussion section: Cannot draft until Results available. Will interpret quantum vs. classical comparison, mechanistic analysis of when/why quantum encoding succeeds or fails, ANP structural class stratification, limitations, and future directions.]

## 5 Conclusion

[PENDING: Conclusion section: Write after Results/Discussion. Summary of findings, implications for ANP-based drug discovery, honest assessment of quantum advantage/equivalence/underperformance, and outlook for hybrid quantum-classical architectures.]

## Supporting Information

Supplementary Information contains: Extended computational details, full ANP metadata tables (ACSI, Fsp<sup>3</sup>, stereocenter counts), quantum circuit diagrams, kernel matrix heatmaps, LOO-CV fold-by-fold predictions, software environment specifications, and data/code availability statement.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability

[PENDING: Data and code availability: GitHub repository URL or Zenodo DOI upon acceptance. Quantum circuit parameters, kernel matrices, and analysis scripts will be made publicly available.]

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